Air University

SSDD (Lab)

Year 2025

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Lab Task # 03

1. **You have to identify the use cases and misuse cases from the given case study. List them in a word document.**

Use Cases (Core Functionalities)

1. Security & Safety Management

→ Detect unauthorized access to the home.
→ Send real-time security alerts to family members.
→ Detect fires and gas leaks.
→ Automatically shut off the gas supply during emergencies.

2. Energy Management

→ Control lighting and HVAC (heating, ventilation, and air conditioning) systems.
→ Optimize energy usage based on occupancy and time.
→ Monitor and analyze energy consumption.

3. Remote Control & Convenience

→ Remotely control home appliances using a smartphone.
→ Automatically adjust lighting, blinds, and curtains.

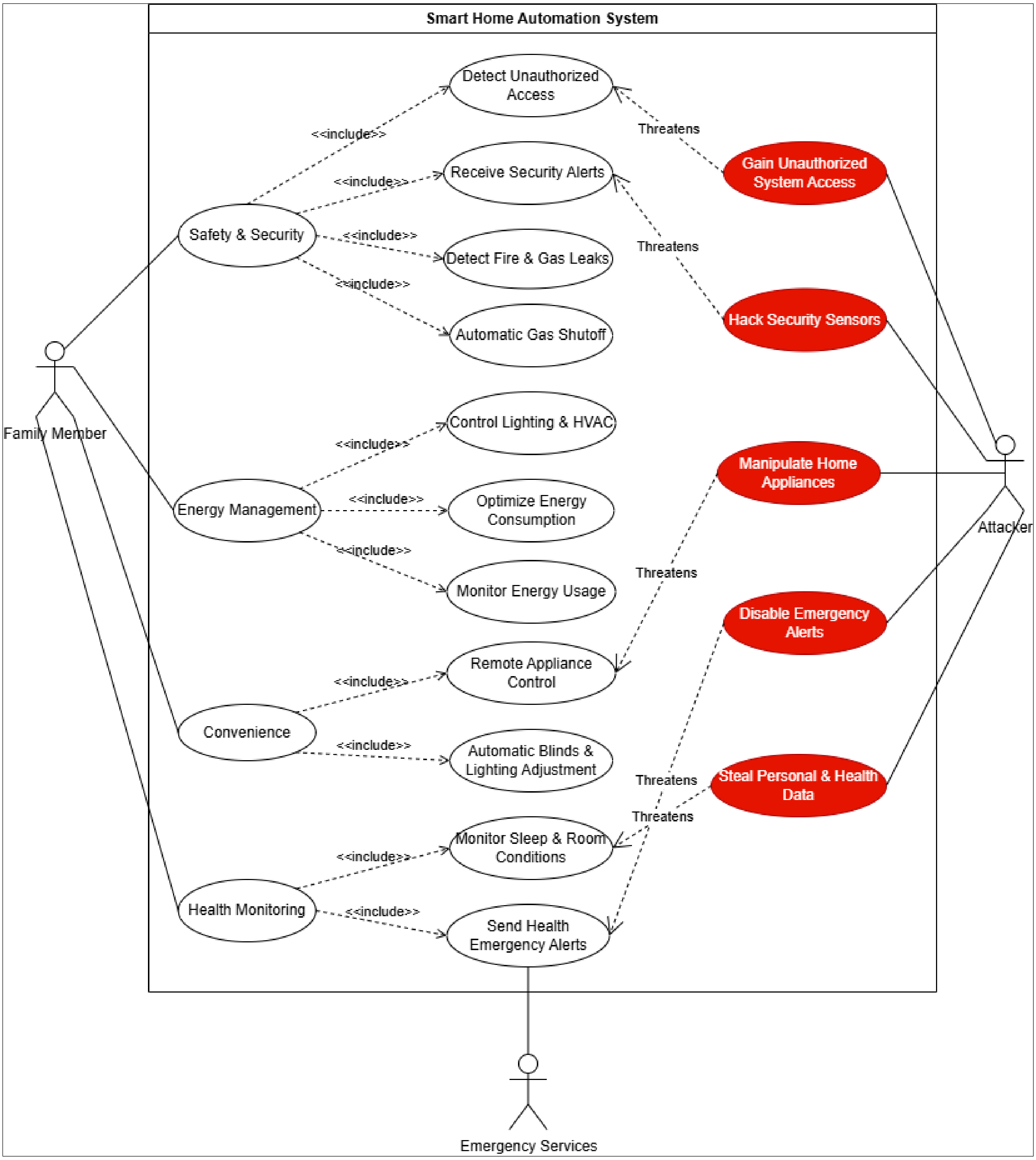
4. Health Monitoring & Emergency Alerts

→ Monitor sleep and room conditions.
→ Send emergency health alerts to predefined contacts.

Misuse Cases (Critical Risks)

→ Unauthorized access to the smart home system.
→ Hacking or manipulation of security devices.
→ Unauthorized remote control of home appliances.
→ False or suppressed emergency alerts.
→ Leakage of personal or health-related data.

2. **Create a UML (Unified Modeling Language) use case diagram to visually represent the interactions between actors (family members, system) and the various use cases and misuse cases. This diagram will help illustrate the functionalities and potential risks associated with the smart home automation system.**



Precautions:

Network Security

- Segment Your Network: Create a separate "Guest" network specifically for your IoT devices. This keeps them isolated from your main computer and phone, preventing hackers from pivoting to more sensitive data if a smart device is compromised.
- Secure Your Router: Change the default username and password for your router immediately. Ensure it is using WPA2 or WPA3 encryption.

Device Security

- Change Default Credentials: Almost all devices ship with a generic default username/password (e.g., "admin"/"password"). Change these immediately to unique, strong passwords.
- Enable Multi-Factor Authentication (MFA): If the device's app supports MFA (requiring a code from your phone in addition to your password), turn it on.
- Update Firmware Regularly: Manufacturers release patches to fix security holes. Enable automatic updates for all devices and apps.

Privacy and Physical Security

- Disable Unnecessary Features: Turn off features you don't use, such as remote access, voice control, or cloud recording, if they aren't essential to the device's function.
- Cover Cameras: Use a physical cover for cameras and microphones when they are not in use, especially in private areas like bedrooms.
- Physical Protection: Ensure outdoor sensors and cameras are installed out of easy reach to prevent tampering.

THE END